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DAY 5

Task 3

Dataset Research Summary – Virtual Try-On Datasets

Introduction

Virtual try-on systems require large and well-structured datasets to train AI models for clothing transfer, body alignment, pose estimation, and outfit visualization. These datasets contain images of people, clothing items, body poses, segmentation masks, and fashion annotations that help deep learning models learn realistic clothing fitting and garment transfer techniques.

The research focused on studying commonly used virtual try-on datasets and understanding their applications in AI-based fashion technology.

DeepFashion Dataset

The DeepFashion dataset is one of the most widely used fashion datasets for computer vision and virtual try-on research. It contains a large collection of clothing images with annotations such as clothing categories, landmarks, textures, and pose variations.

This dataset is mainly used for clothing recognition, fashion recommendation systems, pose estimation, and virtual outfit generation. It provides high-quality fashion images that help AI models learn body-clothing relationships effectively.

The dataset supports tasks such as:

- Clothing detection
- Fashion classification
- Attribute recognition
- Virtual try-on training

VITON Dataset

The VITON dataset was specifically created for virtual try-on applications. It contains images of models along with separate clothing item images that allow AI systems to digitally transfer garments onto human bodies.

This dataset is mainly used in virtual fitting research because it helps models learn:

- Cloth alignment
- Body pose matching
- Garment warping
- Texture preservation

The VITON dataset is commonly used with GAN-based virtual try-on models.

MPV Dataset

The MPV (Multi-Pose Virtual Try-On) dataset contains images of people in different body poses and clothing styles. Unlike standard datasets, MPV focuses on handling pose variations and realistic clothing fitting under different viewing angles.

This dataset improves:

- Pose-aware clothing transfer
- Multi-angle fitting
- Realistic cloth movement simulation

The dataset is useful for training advanced virtual try-on systems that require better pose adaptability.

Dress Code Dataset

The Dress Code dataset is a modern virtual try-on dataset designed for high-quality clothing transfer research. It contains paired clothing images and human images with detailed annotations.

The dataset supports:

- Upper body clothing transfer
- Lower body clothing fitting
- Full outfit generation

It is mainly used in diffusion-based and deep learning virtual try-on systems.

Fashionpedia Dataset

Fashionpedia is a fashion understanding dataset that combines clothing segmentation with detailed fashion annotations. It contains information about clothing styles, accessories, textures, and garment categories.

This dataset is useful for:

- Fashion segmentation
- Clothing parsing
- Fashion recommendation systems
- Attribute detection

Fashionpedia helps AI models better understand clothing details and fashion combinations.

Dataset Comparison

Dataset	Main Purpose	Special Feature
DeepFashion	Fashion analysis	Large fashion annotations
VITON	Virtual try-on	Cloth transfer training
MPV	Multi-pose fitting	Pose variation support

Dataset	Main Purpose	Special Feature
Dress Code	Outfit generation	Detailed clothing transfer
Fashionpedia	Fashion understanding	Rich fashion attributes

Observations

During the research, it was observed that different datasets focus on different aspects of virtual try-on systems. Some datasets are designed for clothing transfer, while others mainly improve pose estimation, segmentation, or fashion understanding.

Datasets with multiple body poses and high-quality annotations produce better virtual fitting results. Proper segmentation masks and clothing alignment data also improve garment transfer accuracy in deep learning models.

Conclusion

Virtual try-on datasets play a major role in training AI-based fashion systems. Datasets such as DeepFashion, VITON, MPV, Dress Code, and Fashionpedia provide important information for clothing fitting, body alignment, segmentation, and outfit generation tasks. These datasets help improve the accuracy and realism of virtual try-on applications and support the development of advanced AI fashion technologies.